

Multiple Choice (10 marks)

1. What is the NMR frequency of ^{31}P in a 20.0 T magnetic field?
 - (a) 54.91 MHz
 - (b) 239.2 MHz
 - (c) 345.0 MHz
 - (d) 2167 MHz
2. Which of the following statements about the lanthanide elements is NOT true?
 - (a) The most common oxidation state for the lanthanide elements is +3.
 - (b) Lanthanide complexes often have high coordination numbers (> 6).
 - (c) All of the lanthanide elements react with aqueous acid to liberate hydrogen.
 - (d) The atomic radii of the lanthanide elements increase across the period from La to Lu.
3. Which of the following statements most accurately explains why the ^1H spectrum of CHCl_3 is a singlet?
 - (a) Both ^{35}Cl and ^{37}Cl have $I = 0$.
 - (b) The hydrogen atom undergoes rapid intermolecular exchange.
 - (c) The molecule is not rigid.
 - (d) Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.
4. Which of the following is always true of a spontaneous process?
 - (a) The process is exothermic.
 - (b) The process does not involve any work.
 - (c) The entropy of the system increases.
 - (d) The total entropy of the system plus surroundings increases.
5. Calculate the equilibrium polarization of ^{13}C nuclei in a 20.0 T magnetic field at 300 K.
 - (a) 10.8×10^{-5}
 - (a) 4.11×10^{-5}
 - (a) 3.43×10^{-5}
 - (a) 1.71×10^{-5}

True / False (10 marks)

Answer True or False

6. True or False: The thermal stability of group-14 hydrides increases in the order: $\text{PbH}_4 < \text{SnH}_4 < \text{GeH}_4 < \text{SiH}_4 < \text{CH}_4$.
7. True or False: In ground-state aluminum, the electron that is most easily removed has an orbital angular momentum quantum number of 2.
8. True or False: The +1 oxidation state is more stable than the +3 oxidation state for aluminum (Al).
9. True or False: The magnetogyric ratio of ^{23}Na is $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$ based on its NMR frequency in a 750 MHz spectrometer.
10. True or False: The infrared absorption frequency difference between DCl and HCl is primarily caused by variations in their dipole moments.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of metal ions in paramagnetic quenching and explain why Cr_3^+ cannot function as a paramagnetic quencher compared to other metal ions.
12. Discuss the significance of B_{12} icosahedra in the solid-state structures of the principal allotropes of elemental boron, including their role in determining physical properties.

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